→ Types of variable:

1) Nominal Data:- categorious, labels with no particular order ex:- Gender, colors, finite values

2) Ordinal:- categories but in order sequence.
ex:- Bechlors, master's phd, customer rating 1 to 5

3) Nomirical:- in numbers:- income age price.
Dnc   cont.

4) categorical:- word categories, can be in Nominal or ordinal.

5) Interval data:- is the type of Nomerical data, b/w two values we get one meaningful information.
ex:- Temp, IQ score. :- in this there is no meaning of true Zero point.

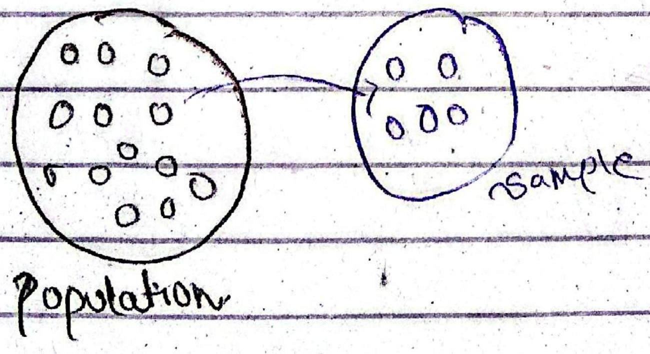
6) Ratio:
same as interval, but have meaning of true value zero. means 0 is impotent.
ex:- height, weight, age,
if we will find out the difference of age b/w ~~2 persons~~ 2persons and if differce will come 0, mean they are in same age.

→ so these all are the types of variable, so in all data sets we see these different kinds of variable.

Population and Sample

Population is the entire group of individuals.
ex: All People in India.
ex2: All Users of netflix
→ so study over population is not possible so.

Sample: we take one group of people from the overall Population and that group is called sample.
so sample is the subset of population.
Note: Sample is best when it represent entire population.
so there are some statical method, through which we can compare that either the sample is related or how much represent to the population



→ why sample?
1. → To reduce the cost of data collection.
2. when a full census data cannot be take.

Sampling:- is a method that allows to choosing a small group from a big group to study it.

we do this because studying everyone is difficult, costly, or takes too much time.

→ what is a Random sample?

A random sample means everyone has an equal chance of bieng selected.

- No favoritism.
- No fixed Pattern
- Totally fair selection

ex:- classroom.

- class has 50 students.
- Teacher wants feedback.
- Teacher randomly picks 10 students (by slips roll nom (lottery))